

Practical Assignmnet 8

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Batch: C14

Write a program that takes one or more file/directory names as command line input and reports following information a) File Type b) Number Of Links c) Time of last Access d) Read ,write and execute permissions.

File:

```
nikhil@macpc:~$ ls  
ass.sh  ass4.sh  file.txt  file2x.txt  filex.txt  linux2025  nik.txt  submission  test.sh
```

Explanation:

Argument Check

- The script checked whether at least **one argument** was provided.
- If no arguments were given, it printed a usage message and exited.

Looping Through Each Input

- It looped through every item passed as an argument ("@\$").
- Each item was treated as a file or directory name.

Existence Check

- For each item, it checked whether the file/directory actually existed.
- If it did not exist, it displayed an error message and skipped to the next item.

File Type Identification

- If the item was a directory, it printed **"File Type: Directory"**.
- If it was a regular file, it printed **"File Type: Regular File"**.
- If it was a symbolic link, it printed **"File Type: Symbolic Link"**.
- Otherwise, it reported the item as a **"Special File"**.

Number of Links

- It used `stat -c %h` to find the number of hard links to the file.
- It then displayed **"Number of Links: <value>"**.

Last Access Time

- The script used `stat -c %x` to extract the last access timestamp.
- It printed this as **"Time of Last Access: <timestamp>"**.

Permission Check

- The script checked read (-r), write (-w), and execute (-x) permissions.
- Based on available permissions, it built a permission string.
- Finally, it displayed the permissions for the file.

Code:

```
#!/bin/bash

if [ $# -eq 0 ]; then
    echo "Usage: $0 file_or_directory_name(s)"
    exit 1
fi

for file in "$@"
do
    if [ ! -e "$file" ]; then
        echo "File/Directory: $file does not exist."
        continue
    fi

    echo "Information for: $file"

    if [ -d "$file" ]; then
        echo "File Type: Directory"
    elif [ -f "$file" ]; then
        echo "File Type: Regular File"
    elif [ -L "$file" ]; then
        echo "File Type: Symbolic Link"
    else
        echo "File Type: Special File"
    fi

    links=$(stat -c %h "$file")
    echo "Number of Links: $links"

    last_access=$(stat -c %x "$file")
    echo "Time of Last Access: $last_access"

    permissions=""
    [ -r "$file" ] && permissions="${permissions}Read "
    [ -w "$file" ] && permissions="${permissions}Write "
    [ -x "$file" ] && permissions="${permissions}Execute"
    echo "Permissions: $permissions"

    echo ""
done
```

Output:

```
nikhil@macpc:~$ vi assignment8.sh
nikhil@macpc:~$ bash assignment8.sh file.txt linux2025
Information for: file.txt
File Type: Regular File
Number of Links: 1
Time of Last Access: 2025-12-04 09:45:34.735502619 +0000
Permissions: Read Write

Information for: linux2025
File Type: Directory
Number of Links: 2
Time of Last Access: 2025-11-29 14:07:27.543304760 +0000
Permissions: Read Write Execute

nikhil@macpc:~$ █
```